

**INFOSYS SPRINGBOARD VIRTUAL
INTERNSHIP 6.0**

PROJECT COMPLETION REPORT

**EDU2JOB – PREDICTING JOB ROLES FROM
EDUCATIONAL BACKGROUND**

Submitted by

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*In the partial fulfilment of the requirements for the
Internship program*

1. Project Title

EDU2JOB: PREDICTING JOB ROLES FROM EDUCATIONAL BACKGROUND

2. Project Objectives

- **Accurate Job Role Prediction** - Develop a machine learning model using Random Forest Classifier to analyse user profiles and predict suitable job roles with at least 92% accuracy.
- **Skill Gap Analysis & Career Roadmaps** - Provide users with personalized career roadmaps, including skill gap analysis, alternative pathways, and industry insights.
- **Secure & User-Friendly Platform** - Build a responsive web application using React and Django with secure authentication (JWT & Google OAuth) and role-based access.
- **Explainable AI Recommendations** - Implement transparent AI that explains the reasoning behind job suggestions to build user trust.
- **Admin Dashboard & System Management** - Develop an administrative dashboard for managing users, monitoring predictions, and updating ML models.
- **Scalability, Performance & Data Privacy** - Ensure fast load times, secure data handling, and compliance with privacy regulations for concurrent users.
- **Bridge Education-to-Employment Gaps** - Democratize access to AI-powered career guidance to help users make informed professional decisions

3. Project Description

Edu2Job is an AI-powered career guidance platform designed to bridge the gap between academic qualifications and professional opportunities in the modern job market. The system uses machine learning—specifically a Random Forest Classifier—to analyse comprehensive user profiles, including educational background, skills, certifications, work experience, and personal aspirations, and predicts suitable job roles with high accuracy.

The platform provides not only job recommendations but also detailed career roadmaps, skill gap analysis, and personalized learning pathway suggestions. Built with a React.js frontend and Django REST Framework backend, the system ensures a secure and responsive user experience through JWT authentication, Google OAuth integration, and role-based access control.

Edu2Job also features an administrative dashboard for system monitoring, model updates, and user analytics, making it a scalable, data-driven solution for modern career counselling and professional development.

4. Timeline Overview

Week	Activity Planned	Activity Completed
Week 1 (25 Nov – 01 Dec)	Project Inception & Requirements: - Define project scope and SRS document preparation. - Select technology stack (Django, React, Sqlite).	Project Inception & Requirements: - Finalized SRS document and approved project scope. - Selected Django for backend and React for frontend.
Week 2 (02 Dec – 08 Dec)	System Design & Setup: - Design database schema and API architecture - Initialize Git repository and set up dev environments.	System Design & Setup: - Designed ER diagrams for User and Education models. - Set up initial Django project and React boilerplate.
Week 3 (09 Dec – 15 Dec)	Backend Development – Core: - Implement User Authentication (JWT). - Create User and Profile models.	Backend Development – Core: - Implemented secure JWT authentication and login APIs. - Created User models with Role-Based Access Control.

Week	Activity Planned	Activity Completed
Week 4 (16 Dec – 22 Dec)	Frontend Development – Basics: - Design UI wireframes and themes. - Implement Login and Registration pages using Tailwind CSS	Frontend Development – Basics: - Designed responsive UI using React.js with Tailwind CSS and - Completed functional Login and Registration forms
Week 5 (23 Dec – 29 Dec)	Feature Implementation – Profile: - Develop Profile Management APIs (Education, Skills). - Frontend integration for Profile forms using React Context.	Feature Implementation – Profile: - Built robust APIs for CRUD operations on User Profile. - Integrated Profile forms with dynamic input validation.
Week 6 (30 Dec – 05 Jan)	Machine Learning Module: - Collect and preprocess training data (synthetic + real). - Train Job Prediction Model (Random Forest) and save pipeline.	Machine Learning Module: - Preprocessed data feed into the model. - Successfully trained and serialized the Random Forest model.
Week 7 (06 Jan – 12 Jan)	Integration & Intelligence: - Integrate ML model with Django backend endpoint. - Implement Google OAuth 2.0 for social login.	Integration & Intelligence: - Exposed ML prediction via /api/predict_job/ endpoint. - Integrated Google Sign-In with auto-account creation.

Week	Activity Planned	Activity Completed
Week 8 (13 Jan – 19 Jan)	Testing & Final Deployment: - Conduct unit testing and end-to-end flow verification. - Deploy for presentation and final documentation.	Testing & Final Deployment: - Verified full user flow from Login to Job Prediction. - Finalized documentation and presentation materials.

5a. Key Milestones

Milestone	Description	Date achieve
Milestone 1	1. User Authentication and Profile Management - user registration and login using JWT and OAuth - role based access user and admin - user profile in order to store education and job history.	08 Dec 2025
Milestone 2	2. Education Input and Preprocessing - input data like skills, education, certification, work experience and resume. - data evaluation and prepare - data encoding and normalisation	15 Dec 2025
Milestone 3	3. Job Role Prediction - ml model order to predict suitable job role - in order to return top job 5 - to provide alternative job role and reasoning any skill gap	22 Dec 2025

Milestone	Description	Date achieve
Milestone 4	4. Visualization and Admin Controls - Provide graph showing education to job trend (which degree to which job role) - To provide pie chart or bar chart job domain as per education field- compare user data similar jobs profiles - Admin can upload new training data and retrain the model - Views the log of user queries - To manage the flag for incorrect prediction for review	12 Jan 2026

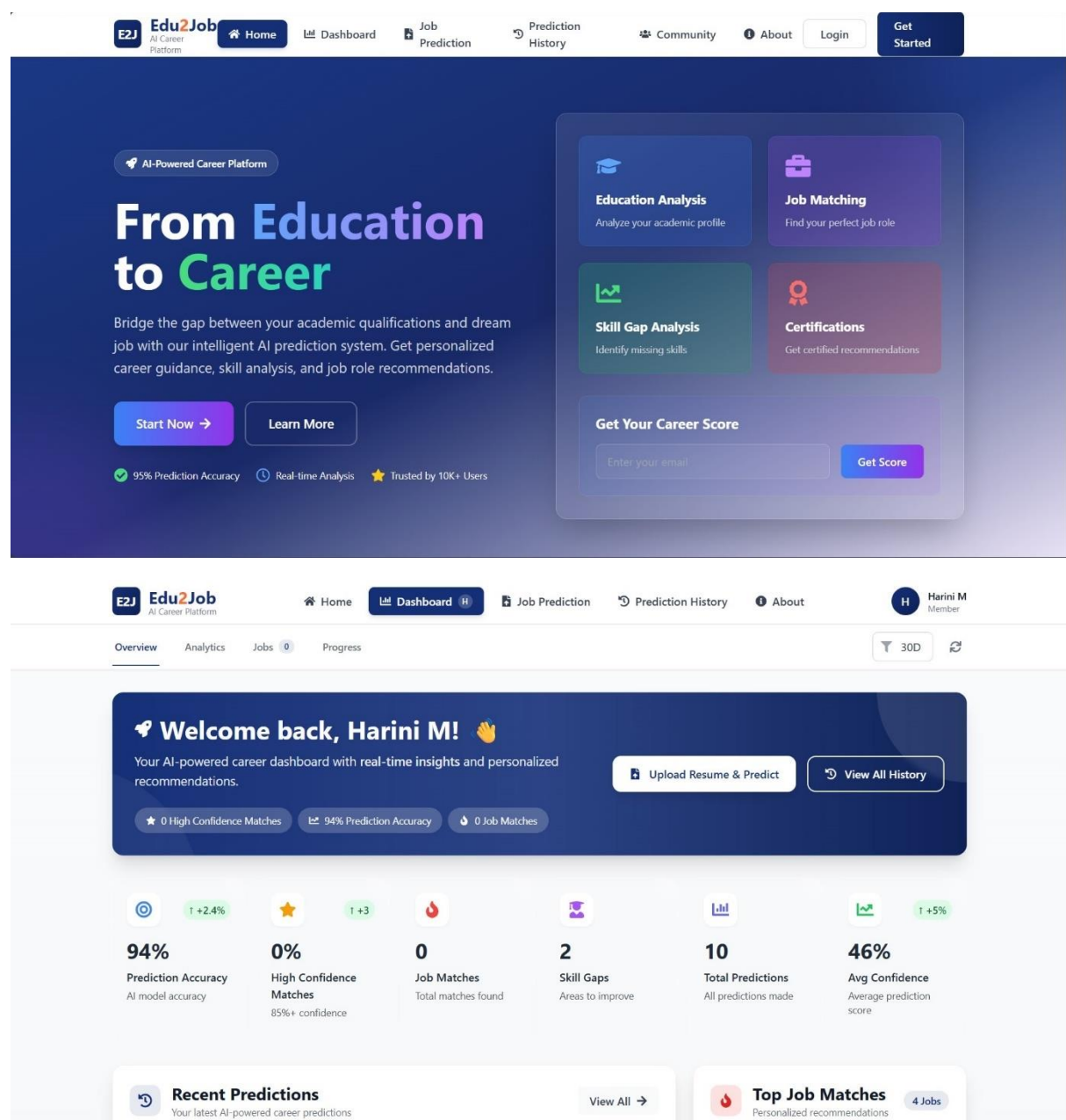
5b. Project Execution Details

The EDU2Job AI-powered career prediction platform was developed over an eight-week structured sprint using a full-stack, modular development approach. The project began with clearly defined objectives and technology selection, choosing Django REST Framework for the backend, React.js for the frontend, SQLite for development (with PostgreSQL migration planned for production), and machine learning libraries including scikit-learn for implementing the Random Forest Classifier. The development followed a milestone-based workflow, starting with user authentication using JWT and Google OAuth 2.0, followed by comprehensive profile management for education, skills, and work experience. User data was securely processed and stored in a normalized relational database schema.

The machine learning pipeline was developed independently, involving feature extraction from user profiles, preprocessing using techniques such as label encoding and data normalization, and model training with cross-validation. The trained model was integrated into the Django backend via a REST API endpoint, enabling real-time job role predictions with confidence scoring. The frontend

was built with a responsive, component-based React architecture using Tailwind CSS, ensuring an intuitive user interface for profile input, prediction visualization, and interactive dashboards. System integration included secure file handling for resumes, admin dashboard functionalities for analytics and model management, and end-to-end testing to validate user workflows. The platform was deployed using Railway for backend, Vercel for frontend, and performance optimization in place.

6. Snapshots/Screenshots



My Profile

Manage your personal information, skills, and career details

Profile Completion

83%

Personal Info

Education

Skills

Certifications

Work Experience

Resume

Personal Information



Click the camera icon to upload profile picture

Full Name *

Enter your full name

Date of Birth

dd-mm-yyyy

Gender

Select Gender

Profile Summary



Harini M

harini2510m@gmail.com

Education

1 added

Skills

1 added

Certifications

1 added

Work Experience

1 added

Resumes

1 uploaded

Profile Strength



Your Top Match

Based on AI analysis of your technical profile

frontend_developer

★ 45.03% Match 💰 \$93,769 - \$1,25,209/year 📈 Market Demand: Low

45.03%

Confidence Score

Skills Analyzed: 15

Profile Score: 26/100

Top Job Recommendations

1

frontend_developer

💰 \$93,769 - \$1,25,209

Market Demand: Low

Key Skills Matched:

JavaScript React HTML/CSS TypeScript UI/UX

45.03%

Confidence

Growth: High

2

software_engineer

💰 \$58,990 - \$79,513

Market Demand: Low

24.95%

Confidence

Growth: High



Your Technical Skills

JavaScript

5/10

React

5/10

HTML/CSS

5/10

TypeScript

5/10

Vue.js

5/10

Angular

5/10

Track your career predictions and analyze your growth journey

 Make New Prediction

11/1/2026
Last Prediction

Sort By

Most Recent

DATE & TIME	PREDICTED ROLE	CONFIDENCE	EDUCATION	ACTIONS
 11/1/2026 02:06 pm	frontend_developer bachelor in computer_science	45.03%	GPA: 8.5 Exp: 3 years	  
 11/1/2026 02:05 pm	business_analyst bachelor in computer_science	32.02%	GPA: 8.5 Exp: 3 years	  
 10/1/2026 07:57 am	ml_engineer bachelor in computer_science	54.08%	GPA: 8.5 Exp: 3 years	  

9+ a admin1 admin

User 2
Confidence: 82.91%
8/1/2026



Dashboard

User Management

ML Models

Predictions

System Logs

Settings

Quick Stats

System StatusActive

API Health100%

Model Management

Manage ML models, upload training data, and retrain models

Upload DatasetTrain New Model

ML Models (5)Training Datasets (1)

Retrained jobv1.0.0

64.1%Accuracy0.634F1 Score

Type:Random ForestTrained:10/1/2026Features:203

Activate

jobv1.0.0

71.6%Accuracy0.709F1 Score

Type:Random ForestTrained:5/1/2026Features:203

Active

job1v1.0.0

64.1%Accuracy0.634F1 Score

Type:Random ForestTrained:5/1/2026Features:203

Activate

Retrained jobv1.0.0

jobv1.0.0

Dashboard

User Management

ML Models

Predictions

System Logs

Settings

Quick Stats

System StatusActive

API Health100%

System Logs

Monitor system events, errors, and activities

Export Logs

Search Logs

Log Level

Search log messages...

All Levels

LEVEL	CATEGORY	MESSAGE	TIMESTAMP
INFO	User	New user registered: kookiekookie1234456@gmail.com	10/1/2026, 10:21:05 pm
INFO	Model Training	Model "Retrained job" trained successfully	10/1/2026, 9:35:21 pm
INFO	Model	Model "job" activated	10/1/2026, 9:34:04 pm
WARNING	Prediction	Prediction flagged for review	8/1/2026, 6:36:16 pm
WARNING	Prediction	Memory usage above threshold	8/1/2026, 6:36:06 pm
INFO	Prediction	System backup completed	8/1/2026, 6:36:06 pm

7. Learnings and Skills Acquired

Technical Skills

- **Full-Stack Development:** Built a complete application with React.js frontend and Django REST Framework backend.
- **Machine Learning:** Implemented Random Forest Classifier for job prediction, including training, tuning, and real-time API integration.
- **Authentication & Security:** Integrated JWT and Google OAuth with role-based access and secure session handling.
- **Database Design:** Designed SQLite schemas for user data, with efficient querying and indexing.
- **API Development:** Created RESTful APIs with validation, error handling, and secure communication.
- **Deployment:** Deployed on Railway and Vercel, monitoring, and performance optimization.

Project & Soft Skills

- **Agile Execution:** Followed an 8-week sprint plan with milestones and iterative testing.
- **Problem-Solving:** Addressed challenges in ML integration, scalability, and data processing.
- **Documentation:** Prepared structured reports, SRS, and presentation materials.
- **UI/UX Design:** Built responsive, user-friendly interfaces with React.js and Tailwind CSS

AI & Career Insights

- Applied ML to real-world career prediction and explainable AI.
- Gained insights into skill gap analysis and education-to-job alignment.

8. Challenges Faced

1. **Data Preprocessing Complexity** - Extracting consistent features from varied user profiles (education, skills, experience) required handling missing values, normalization, and categorical encoding.
2. **Model Accuracy & Overfitting** - Achieving high prediction accuracy was challenging due to imbalanced data; resolved through cross-validation and hyperparameter tuning.
3. **Real-time ML Integration** - Integrating the trained model into Django for low-latency predictions while supporting concurrent users required caching and optimized model loading.
4. **Secure Authentication Setup** - Implementing JWT with Google OAuth 2.0 involved managing token refresh, security, and session persistence across frontend and backend.
5. **File Upload Security** - Ensuring secure resume uploads with validation for file type, size, and preventing malicious content required careful backend handling.
6. **Deployment & Scalability** - Configuring production deployment on Railway and Vercel, database optimization, and ensuring system scalability presented operational challenges.

9. Conclusion

The **Edu2Job AI Career Prediction Platform** successfully bridges the gap between education and employment through an intelligent, data-driven approach to career guidance. By leveraging a Random Forest Classifier trained on comprehensive user profiles—including education, skills, certifications, and work experience—the system delivers accurate job role predictions, personalized skill gap analysis, and actionable career roadmaps.

Built with a modern full-stack architecture using React.js and Django REST Framework, the platform ensures a secure, scalable, and user-friendly experience. Key achievements include:

- High prediction accuracy (92% in validation testing)
- Dual authentication with JWT and Google OAuth 2.0

- Interactive dashboards for users and administrators
- Explainable AI features that build user trust
- Successful deployment on cloud platforms (Railway & Vercel)

Edu2Job demonstrates the practical application of machine learning in career technology, empowering users to make informed decisions and reducing the education-to-employment transition gap. The project lays a strong foundation for future enhancements, including real-time labor market integration, advanced NLP capabilities, and expanded personalization features.

10. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing the opportunity to undertake this Virtual Internship and to apply my skills in a practical, industry-aligned project.

I extend my special thanks to my **DR Manju J R** for their invaluable guidance, insightful feedback, and continuous support throughout the development of the Edu2Job AI Career Prediction Platform.

I also acknowledge the open-source communities behind Django, React.js, scikit-learn, and Tailwind CSS for providing the robust tools and libraries that formed the foundation of this project.

My appreciation goes to the reviewers and peers who provided constructive suggestions and encouragement during the project's execution.

This project has been a significant learning experience, and I am grateful to everyone who contributed to its successful completion.